

100 Days of Linux - Day 054

Title: Secure Remote Access with SSH

Today's Goal

Learn how to connect securely to remote Linux systems using the **ssh** command and understand the basics of SSH authentication.

Why This Matters

SSH is the standard way to manage Linux servers in cloud platforms, DevOps, and data engineering. It allows you to execute commands on remote machines securely.

Command of the Day: ssh

Common Syntax:

```
ssh username@hostname
ssh username@IP_ADDRESS
ssh -p 2222 username@hostname (custom port)
ssh -i ~/.ssh/id_rsa username@hostname (private key)
```

Common Options:

- p = Specify SSH port
- i = Use a specific private key
- v = Verbose troubleshooting output

Authentication:

- Password authentication
- SSH key authentication (recommended)

Beginner Lesson

SSH creates an encrypted connection between your computer and another Linux machine. Most cloud servers are managed almost entirely through SSH. Even if you don't have a remote server today, it's important to understand the command and workflow.

Practice Questions

1. Run 'ssh' without arguments and read the help message.
2. Check whether your system already has an SSH key by listing the contents of ~/.ssh.
3. Display your public key if one exists. If not, note that no key is available yet.
4. Write the command you would use to connect to a server with username ubuntu at IP address 192.168.1.100.
5. Display your current working directory after completing today's tasks.

Hints

You do not need access to a real server today. Focus on understanding the command syntax and checking your local SSH configuration.

Mini Challenge

Create a folder named `ssh_practice` and a text file named `servers.txt` containing three imaginary server addresses. Write the SSH command you would use to connect to each one.

Common Mistakes

Sharing your private SSH key. Never send your private key to anyone. Only your public key should be shared when required.

Did You Know?

GitHub, cloud providers, and many Linux servers support SSH key authentication because it is generally more secure than passwords.

Pro Tip

Most professional developers create an SSH key once and reuse it securely across trusted servers and Git hosting services.

Answer Key (Instructor Only)

Q1: `ssh`

Q2: `ls -la ~/.ssh`

Q3: `cat ~/.ssh/id_rsa.pub` (or `~/.ssh/id_ed25519.pub` if available)

Q4: `ssh ubuntu@192.168.1.100`

Q5: `pwd`

Homework

Explore the `~/.ssh` directory and read about SSH keys. If you already have a GitHub account, check whether you've added a public SSH key.

Next Day Preview

Tomorrow you'll learn SCP to securely copy files between Linux systems using SSH.